

Cornell Electric Vehicles – Mechanical

Coast By Wire

Spring 26 Technical Report

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Executive Summary

- 1) What are the desired function(s) of your design?
 - a) The purpose of this project is to investigate new ways to allow Cornell Electric Vehicles' (CEV) Urban Concept car to drive backwards, without sacrificing its ability to coast. For reference, CEV is a project team dedicated to building hyper-efficient autonomous electric vehicles. Each year, we build a car to compete in the Shell Eco-marathon, a competition that pushes high school and college students to build the most efficient car possible. To be competitive, most teams use one-way bearings, which allow their vehicles to coast like a bicycle. While these bearings are great for maximizing efficiency, they create a massive issue for our autonomy initiative. In this configuration, the car can only drive forward, which is extremely inconvenient for navigating in small spaces or parking. More specifically, this project will investigate the use of a controllable electromagnetic (EM) clutch as a replacement for a one-way bearing. The final product is a benchtop testing rig design that can be used to validate custom controls. These controls will allow the car to be in one of three modes: forward lock, forward coasting, or reverse lock. In forward lock and reverse lock driving modes, the motor is always connected to the wheels. In coasting mode, the clutch behaves like a one-way bearing.
- 2) What constraints related to the main function(s) must your design satisfy?
 - a) The clutch must be actuated automatically based on control logic. It should require no input from the driver, aside from selecting the driving mode. This is primarily a risk mitigation strategy; it lowers the likelihood of equipment damage and also increases the safety of the driver by reducing additional distractions.
 - b) The system must allow bidirectional driving (forward and reverse).
 - c) The system's control logic must allow the vehicle to coast. Coasting is essential to maximizing efficiency.
 - d) The benchtop testing rig must have RPM sensors on both sides of the clutch. The motor speed and the wheel speed are important parameters that are used to determine if the clutch should be engaged or not.
 - e) The design also needs to be compatible with next year's motor selection. This means the clutch selected must be able to handle a speed of 2613.49 RPM and a torque of 18.11 N*m, based on vehicle dynamics calculations done earlier this semester (Christina Ge).
- 3) What are the performance objectives of your design? (Give quantitative metrics as much as possible).
 - a) The system consisting of the clutch and the clutch mounting plate must not decrease the theoretical efficiency of the car by more than 10%.
 - b) The control logic should prevent torque spikes.
- 4) What alternative design concepts were considered?
 - a) Many other designs were considered for this project. Brief descriptions of these ideas are outlined below.

Software Enhanced Mechanical Clutch

Mechanical clutches, primarily dog clutches are great for torque transmission with zero slip. Much like the selected design, the clutch would be placed in between the motor and the wheel and be actuated following the logic of a one-way bearing. This design presents added challenges, though, as it requires an additional mechanism to actuate the clutch, most likely an extra motor. However, because it is a mechanical design, there is more flexibility to implement something more complicated, like bi-stable mechanisms. A bi-stable mechanism would decrease the power draw of this system by only requiring power to turn the clutch on and off.

Modular Wheels

Another proposed design was to simply have two sets of wheels for the car. One set would be equipped with one-way bearing for optimized efficiency. The other would have regular bearing to allow power transmission in both directions, ideal for an autonomous demonstration. This design is by far the simplest, requiring no extra parts on the vehicle. However, it fundamentally goes against the goals of this project: to provide a solution to allow coasting and bidirectional driving in the same vehicle.

Double Clutch Transmission

This design pulls from standard manual transmissions and uses intermediate shafts to add a reverse gear. The power from the motor is split into two shafts using a belt and sprocket system. The top shaft uses a one-way clutch to handle forward acceleration and coasting. The bottom shaft uses an EM clutch to connect to an additional shaft fitted with a set of gears that reverses the direction of the wheels. This design is much more complex than the others, but it does provide both coasting and reverse drive without the use of any advanced controls or software. The EM clutch can be triggered with something as simple as a switch. See **Figure 1**.

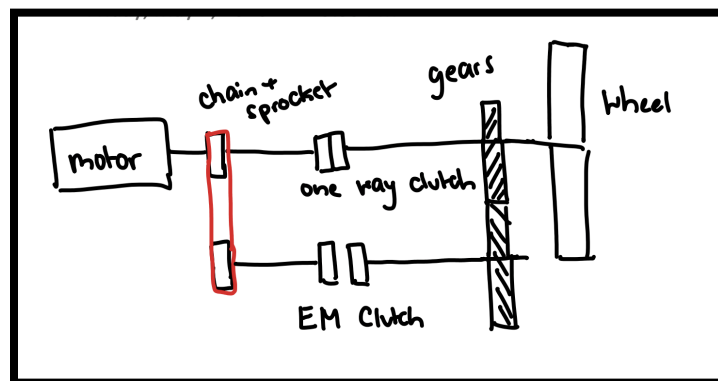


Figure 1. Sketch of Double Clutch Transmission

- 5) What analyses were used to select among these alternative design concepts?
- a) A decision matrix was created to evaluate the efficacy of all design concepts. The factors in this matrix are weight, mechanical complexity, power draw, software complexity, time to switch between modes, and space. Each factor received a weight, which was based on how much it would affect the efficiency of the car. For example, weight (mass) was worth 20% of the total score. This is because a large change in weight significantly increases the energy lost in both acceleration and coasting. On the other hand, software complexity was weighted only 5%, as it does not directly decrease the efficiency of the car. It should also be noted that switching time was weighted 40%, since it embodies the main goal of this project. See **Table 1**.

Table 1. Decision Matrix

Category Weights	0.2	0.1	0.1	0.05	0.4	0.15	1
Designs	Weight (lowest)	Mechanical Complexity (lowest)	Power Draw (lowest)	Software Complexity (lowest)	Time to switch between modes (lowest)	Space (lowest)	Total
Software Enhanced Mechanical Clutch	3	3	3	2	5	3	3.75
Software Enhanced EM Clutch	4	4	4	2	5	5	4.45
Modular Wheels	5	5	5	5	1	5	3.4
Double Clutch Transmission	1	1	5	5	5	1	3.2

- 6) What industry or society standards were used to inform or evaluate your design?
- a) This project started with extensive market research to see how this problem is solved in industrial automotive settings. This led to the discovery of an Overrunning Dynamic Controllable Clutch (ODCC). Depending on the manufacturer, this type of clutch can switch between 4 different operating modes, including coasting and direct drive in both directions. However, these clutches are usually designed for much larger applications, like full sized trucks. For this project, it did not make sense to buy one off the shelf, as it would be incredibly overspec'd. However, the idea of putting all that functionality into one small device was highly influential for this project. It informed the decision to use an EM clutch over a mechanical one and demonstrated that one-way (freewheeling) behavior can be achieved through software.
- 7) Which concepts or skills learned in your coursework were applied to the design? Projects are expected to make substantial use of MAE and related ENGRD classes. Please provide a list with each entry providing the department and number of the course, plus a brief description of the particular concept or skill used.
- a) MAE 2250 (Introduction to Mechanical Design): Applied CAD fundamentals to create the testing rig assembly in Autodesk Inventor.
- b) MAE 2030 (Dynamics): Applied energy methods to determine vehicle energy lost in acceleration, rolling resistance, and aerodynamic drag. Turned these concepts into a spreadsheet to predict how a small increase in mass will affect the overall efficiency, based on data collected at this past competition.
- c) MAE 3260 (System Dynamics): Applied knowledge of system block diagrams by using Simulink to simulate the behavior of the clutch. Directly used modelling concepts from class to create a model of the motor-wheel system.

- d) MAE 3780 (Mechatronics): Applied basic equations from class to calculate the power draw of the clutch over the course of the race.
- 8) Evaluate your design, relative to its function(s) and constraints. How well did your design meet each of the performance objectives? How well does your design compare to other, existing solutions to the problem?
- a) The design met all of the functional requirements and constraints, while only decreasing the overall efficiency of the vehicle by 3.96%. Additionally torque spikes were mainly eliminated in the Simulink model. This is a great accomplishment for CEV since nothing like this has been done in the past. Compared to industrial solutions, like the Amsted Automotive Dynamics Controllable Clutch, this design is slightly less efficient, as its synchronization software is not nearly as advanced. This means that every time the clutch engages, some energy is lost to slip. The power consumption is also slightly higher since this design uses an EM clutch, which draws power while engaged, instead of a bistable design, which only draws power while switching states.
- 9) What impact do you see your design, if implemented, having upon public health, safety, and human welfare, as well as upon current global, cultural, social, environmental, and economic concerns?
- a) If implemented, this design would be revolutionary for CEV. With the ability to reverse, the autonomy team can begin to develop much more advanced navigation algorithms. Beyond the team, technology like this will help increase efficiency and add functionality to many smaller electric drivetrains, like those found in manufacturing and robotics. More efficient drivetrains like these decrease the demand for energy, decreasing the cost of operation.
- 10) What format did your design take? For example, is it a complete set of CAD drawings, a working prototype, a full finished product, a system configuration, a process map, or something else?
- a) The design is a full CAD assembly of a testing rig that can be used to verify the chosen clutch and controls. The report also includes a manufacturing plan and BOM. Lastly, a Simulink model is presented to demonstrate the desired behavior of the clutch.
- 11) Describe each student's role in the design project if it was a group project.
- a) This project was completed independently.

System Description

The purpose of Coast by Wire is to allow future Urban Concept cars to drive backwards without sacrificing the ability to coast. This is achieved by controlling an off the shelf electromagnetic clutch with custom software, such that it behaves like a one-way bearing. Ideally, the system interfaces with the drivetrain of the car, placed in between the motor and the driveshaft. However, for this project, the focus was on researching and testing the technology, rather than implementing it in next year's car. This project has two main parts: the control logic and the benchtop testing rig. The control logic is presented mainly through Simulink. The benchtop is a full CAD model designed to test this software, under load, but not actually in the car. **See Figure 2.**

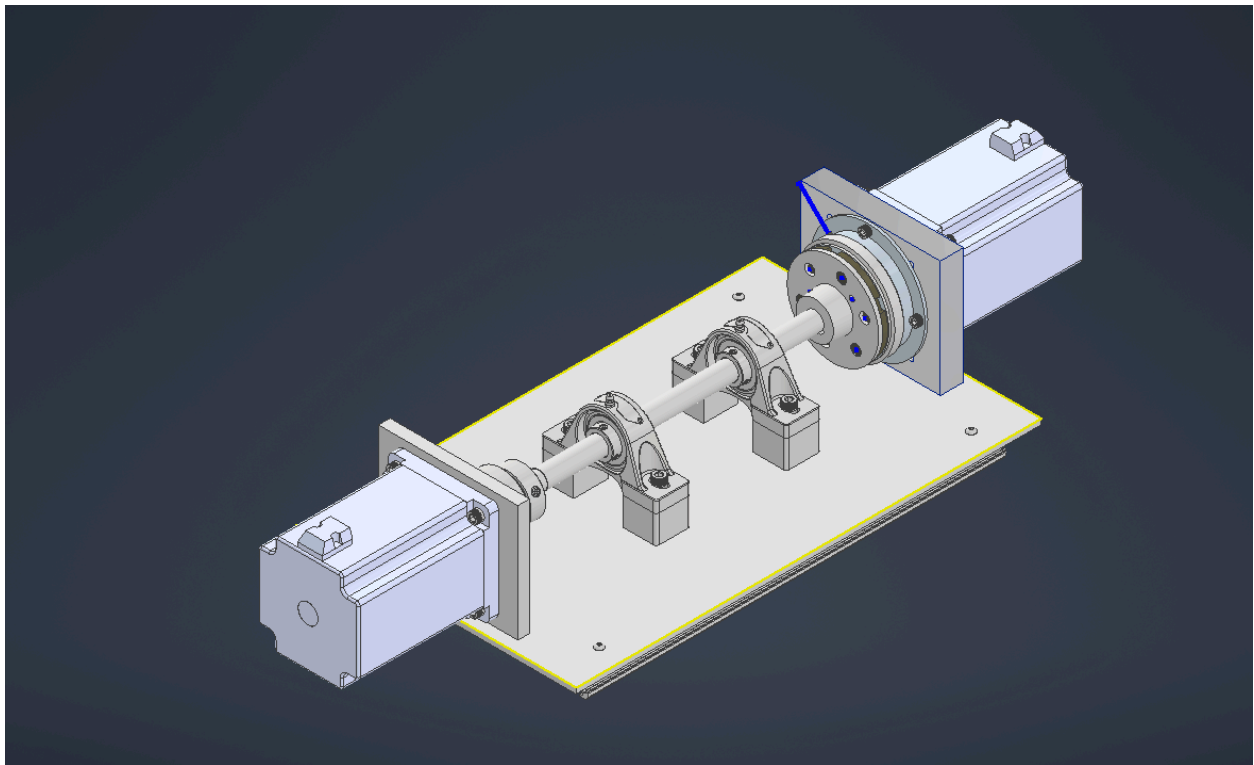


Figure 2. Full Benchtop CAD Assembly

Terminology

- Electromagnetic (EM) Clutch- a device that is used to couple two shafts on command using an electromagnet that is turned on by applying a current.
- Stator- The part of the clutch that does not move. It generates the magnetic field.
- Rotor- The moving part of the clutch.
- Armature- The part of the clutch "engages" when a magnetic field is applied.
- One way bearing- A type of bearing that only allows torque transmission in one direction. Used to coast or freewheel.
- Coast, Freewheel- A state where the wheels are spinning and the car is moving forwards, but the motor is not drawing any power.
- Bi-Stable Mechanism- A mechanism that has two equilibrium states. No force is required to keep the mechanism in one of those states. Energy is only required to switch between them.

Research and Requirements

Background Research

UC '25 Architecture

To better understand this project, one needs to understand the current drivetrain architecture. See **Figure 3**. The design for UC'26 uses a one-wheel drive system. The output shaft of the gearbox is coupled to the driveshaft via a flexible shaft coupling. This adds compliance to the system to protect from vibrations and slight misalignments. The wheel hubs slide onto the driveshaft. One way bearings are pressed into the wheel hubs, such that the motor can drive the wheels, but the wheels can never drive the motor. See **Figure 5**. This behavior is called “freewheeling” or “coasting”, and is common in bicycles. During competition runs, these bearings are extremely helpful for maximizing efficiency, since they allow the vehicle to move forward without using any power. However, in this current configuration, it is impossible to drive the car backwards. Spinning the motor in reverse has no effect on the wheels.

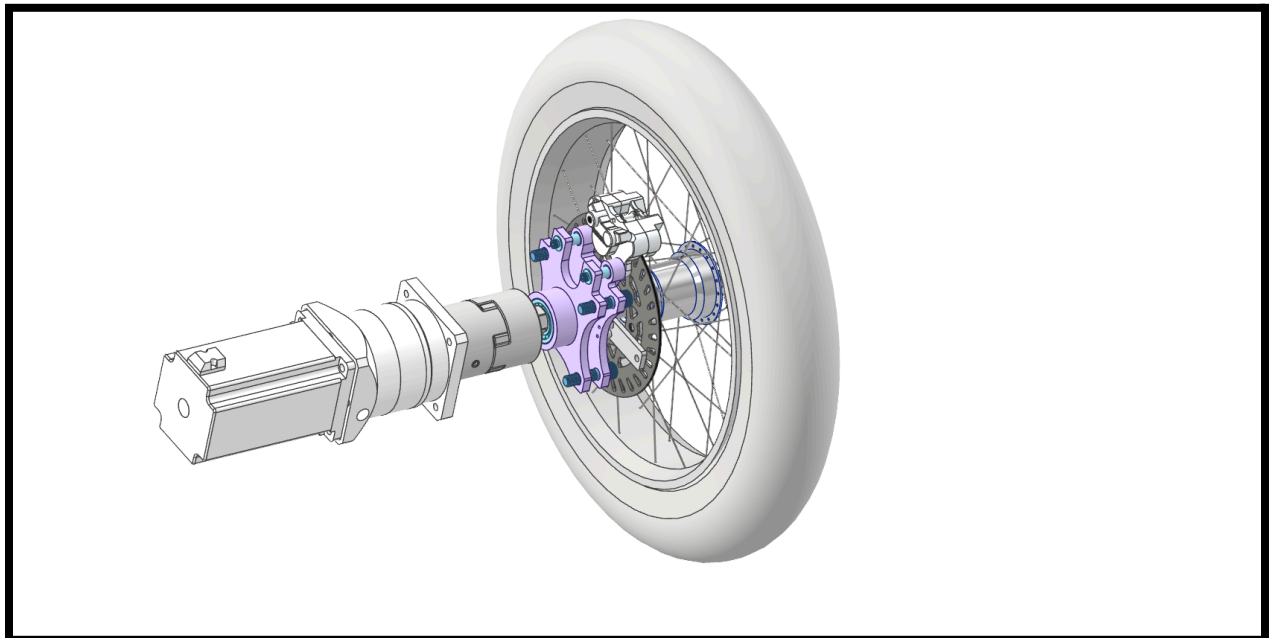


Figure 3. UC '26 Drivetrain Full Assembly

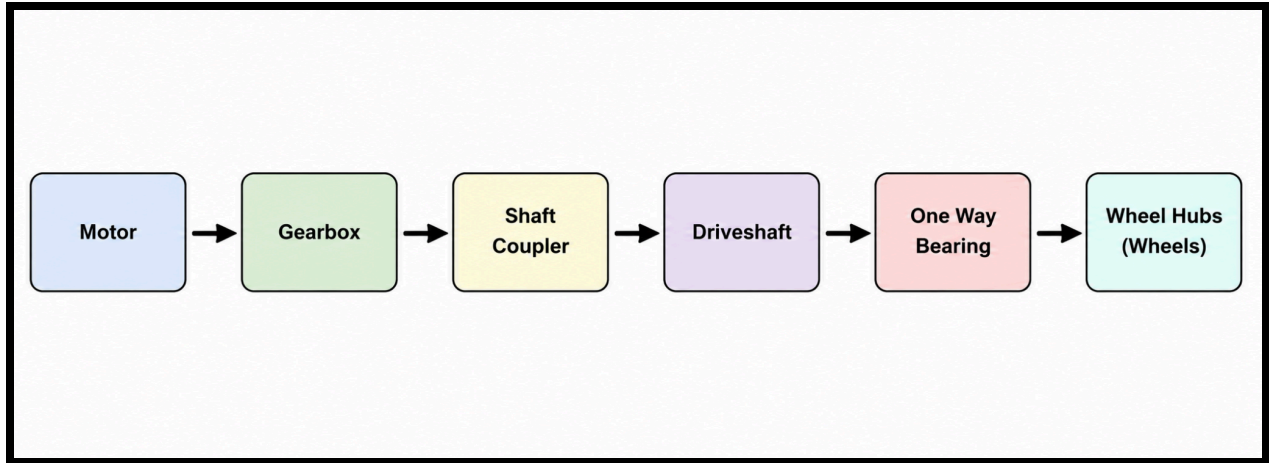


Figure 4. Power transmission diagram

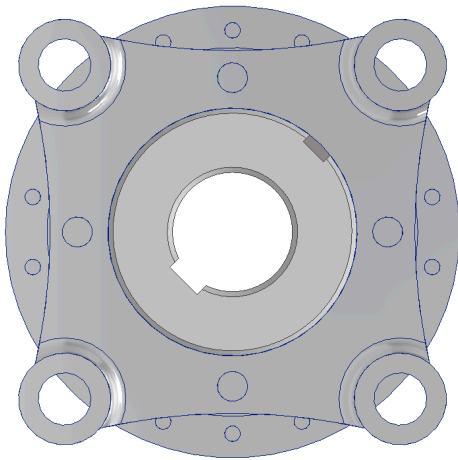


Figure 5. Wheel Hub with bearing press fit inside

Clutch Landscape

An understanding of clutches is also necessary to understand this project. A clutch is a device, commonly used in automobiles and heavy machinery, that is used to selectively couple two shafts. Most clutches can be categorized by their actuation type and torque transmission type. For example, a dog clutch transfers torque mechanically through interlocking teeth. However, it can be actuated manually or electromechanically through a small motor. The clutch used in this report is electromagnetic. This means that it is actuated by an electromagnetic force, but the torque transfer is still mechanical, in this case, friction.

For this report, two main types of clutch were considered: an EM clutch and a dog clutch. A comparison of the benefits and drawbacks is included in the table below.

EM Clutch 	Dog Clutch 
Pros: • Easily actuated automatically • Allows some slip, simpler software	Pros: • Teeth prevent slipping, higher efficiency (with proper meshing) • If used with a bistable mechanism, only requires power when switching states
Cons: • Slip leads to efficiency losses • Requires additional power while engaged	Cons: • Requires more complicated software to avoid improper meshing and damaging hardware • Needs secondary mechanism to actuate

Why Use a Clutch?

A one way bearing is essentially a type of passive shaft coupler; it couples the spinning of two different bodies, but only in one direction. More specifically, it follows the logic.

$$\text{State} = \begin{cases} \text{Engaged,} & \omega_{\text{motor}} > \omega_{\text{wheel}} \\ \text{Disengaged,} & \omega_{\text{motor}} < \omega_{\text{wheel}} \end{cases}$$

The issue with a one way bearing is that it is hard coded to behave this way since it is a purely mechanical device. A clutch is also a type of shaft coupler, but it is active. This means that its behavior is not predetermined. Depending on the type of clutch, software can actually be written to control the clutch. So unlike a one way bearing, a clutch can switch between multiple different behaviors, for example coasting and driving backwards.

Design Requirements

The clutch must be actuated automatically based on control logic and sensor inputs. It should require no input from the driver, aside from selecting the driving mode (forward-lock, forward coasting, or reverse-lock), which will tell the clutch which control logic to use. This is primarily a risk mitigation strategy; it lowers the likelihood of equipment damage and also increases the safety of the driver by reducing additional distractions. An automatic clutch is also much more efficient, as a computer can react much faster than a human driver.

The system must allow bidirectional driving (forward and reverse). This is the primary goal of the project.

The system's control logic must allow the vehicle to coast. Coasting is essential to maximizing efficiency.

The benchtop testing rig must have RPM sensors on both sides of the clutch. The motor speed and the wheel speed are important parameters that are used to determine if the clutch should be engaged or not.

The design also needs to be compatible with next year's motor selection. This means the clutch selected must be able to handle an rpm of 2613.49 and a torque of 18.11 Nm, based on vehicle dynamics calculations done earlier this semester (Christina Ge).

Key Parameters

The system consisting of the clutch and the clutch mounting plate must not decrease the theoretical efficiency of the car by more than 10%. These are the parts of the assembly that would actually go into the car if this design was incorporated. 10% corresponds to 3.7 mi/kWh for the team's current efficiency. This is an extremely small difference. For reference, at the 2026 competition, an increase of more than 10% in efficiency was achieved by simply driving better. This is because a large portion of energy is lost in the 7 mandatory stops that the vehicle must make throughout the 10 mile race. Timing the deceleration of the car, such that the driver barely has to use the brakes, can make a huge difference in our efficiency.

When engaged, clutch torque is proportional to the difference in velocity of the motor and the wheel, as described in the equation below. If the motor is spinning much faster than the wheel, this can lead to large spikes in torque. Limiting these spikes below the torque rating of the clutch (20 Nm) is essential to prevent slipping and loss of energy.

$$\tau_c = K_c \Delta\omega$$

Potential Designs

Software Enhanced EM Clutch

In this design, software is used to control an electromagnetic clutch. This is an extremely simple configuration since the EM clutch requires no additional actuation mechanism. All that is required is a small adapter plate to attach the clutch to the motor. However, because of the way an EM clutch works, it draws power constantly while engaged.

Software Enhanced Mechanical Clutch

Mechanical clutches, primarily dog clutches are great for torque transmission with zero slip. Much like the selected design, the clutch would be placed in between the motor and the wheel and be actuated following the logic of a one-way bearing. This design presents added challenges, though, as it requires an additional mechanism to actuate the clutch, most likely an extra motor. However, because it is a mechanical design, there is more flexibility to implement something more complicated, like bi-stable mechanisms. A bi-stable mechanism would decrease the power draw of this system by only requiring power to turn the clutch on and off.

Modular Wheels

Another proposed design was to simply have two sets of wheels for the car. One set would be equipped with one-way bearing for optimized efficiency. The other would have regular bearing to allow power transmission in both directions, ideal for an autonomous demonstration. This design is by far the simplest, requiring no extra parts on the vehicle. However, it fundamentally goes against the goals of this project: to provide a solution to allow coasting and bidirectional driving in the same vehicle.

Double Clutch Transmission

This design pulls from standard manual transmissions and uses intermediate shafts to add a reverse gear. The power from the motor is split into two shafts using a belt and sprocket system. The top shaft uses a one-way clutch to handle forward acceleration and coasting. The bottom shaft uses an EM clutch to connect to an additional shaft fitted with a set of gears that reverses the direction of the wheels. This design is much more complex than the others, but it does provide both coasting and reverse drive without the use of any advanced controls or software. The EM clutch can be triggered with something as simple as a switch. See **Figure 6**.

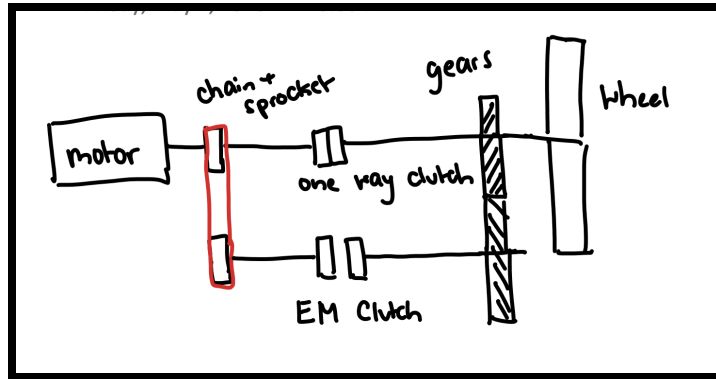


Figure 6. Sketch of Double Clutch Transmission

Selected Design and Justification

A decision matrix was created to evaluate the efficacy of all design concepts. The factors in this matrix are weight, mechanical complexity, power draw, software complexity, time to switch between modes, and space. Each factor received a weight, which was based on how much it would affect the efficiency of the car. For example, weight (mass) was worth 20% of the total score. This is because a large change in weight significantly increases the energy lost in both acceleration and coasting. On the other hand, software complexity was weighted only 5%, as it does not directly decrease the efficiency of the car. It should also be noted that switching time was weighted 40%, since it embodies the main goal of this project. Using this matrix, the Software Enhanced EM Clutch was chosen as the design with the most potential. See **Table 2**.

Category Weights	0.2	0.1	0.1	0.05	0.4	0.15	1
Designs	Weight (lowest)	Mechanical Complexity (lowest)	Power Draw (lowest)	Software Complexity (lowest)	Time to switch between modes (lowest)	Space (lowest)	Total
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Software Enhanced EM Clutch	4	4	4	2	5	5	4.45
Modular Wheels	5	5	5	5	1	5	3.4
Double Clutch Transmission	1	1	5	5	5	1	3.2

Table 2. Decision Matrix

Detailed Design (CAD)

Clutch Selection

The [101-10-11G-24V EM Clutch from Miki Pulley](#) was selected for this design. It is rated for 20 Nm of torque and 5000 RPM, which satisfies the torque and RPM requirements set by next year's motor. It draws 20 W of power and weighs only 1.660 kg. CAD of this clutch was also available, which made this the perfect choice for this project. Many other clutches from the same manufacturer were considered, specifically [101-20-11G-24V](#), which has specifications of 160 N*m, 2500 RPM, 45W, and 10.5kg. This clutch would be ideal for placing after the gearbox instead of before, although it is much heavier and draws much more power. Normally, placing a clutch between the motor and gearbox would be unnecessarily complicated, but since the team is transitioning to a chain and sprocket system, it is much more viable now.



Benchtop Design Overview

The benchtop uses 2 motors, arranged such that they drive the same shaft. **See Figure 7.** The motor on the right drives the whole assembly. The motor on the left acts as the loading for this benchtop. By shorting the wires together, the motor provides a resistive load that is proportional to the angular velocity of the shaft. Alternatively, the motor can be controlled via software to output a constant torque. An additional motor was selected for the load to make this setup highly customizable. With a motor controller, this benchtop can simulate any number of loading conditions, like uphill driving or a flat race track. Additionally, both motors have built in Hall sensors to measure the RPM which is essential for

the one-way logic.

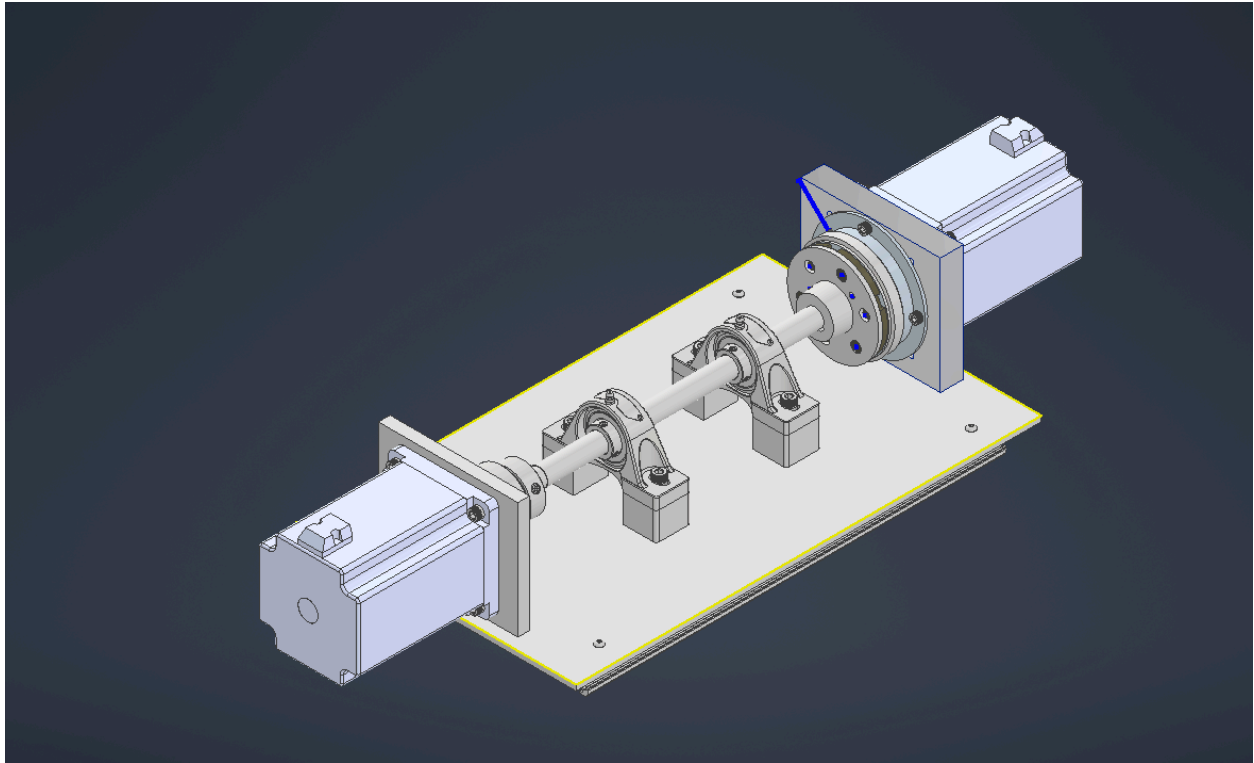


Figure 7. Benchtop Full Assembly View 1

Clutch Plate

Both the driving motor and the clutch are mounted to the clutch plate. See **Figure 8**. To accomplish this, the plate has threaded mounting holes for both off the shelf parts. The clutch mounting holes are threaded to avoid interference with the top of the motor. The motor mounting holes are threaded to aid in assembly and disassembly. For this design, it was important that the motor can be removed easily for on-car testing. The clutch plate also has 4 threaded mounting holes on the bottom to bolt it to the larger benchtop mounting plate below. See **Figure 9**. Lastly, a large hole is cut out in the middle of the part to decrease weight and to allow the plate to sit flat on the motor, which has a raised circular profile.

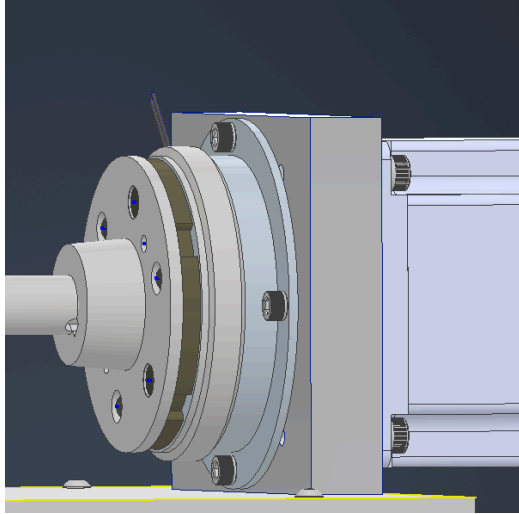


Figure 8. Clutch Plate with Motor and Clutch attached

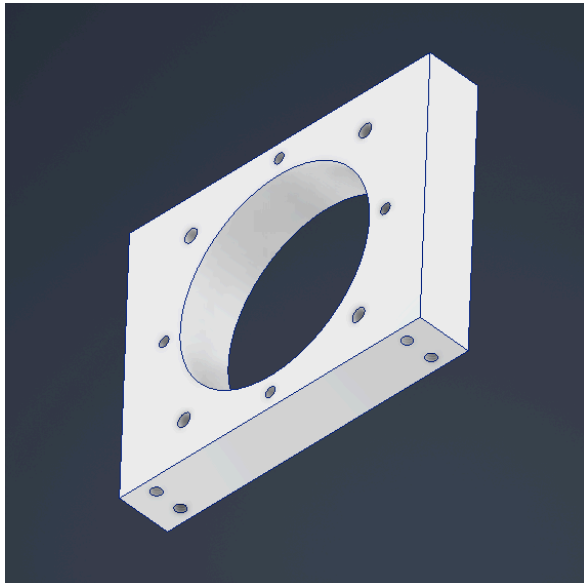


Figure 9. Clutch Plate

Motor Plate

The load motor is mounted similarly to the drive motor. **See Figure 10.** Just like the clutch plate, the motor plate has threaded mounting holes for the motor. It also has threaded holes to be bolted to the mounting plate below. **See Figure 11.**

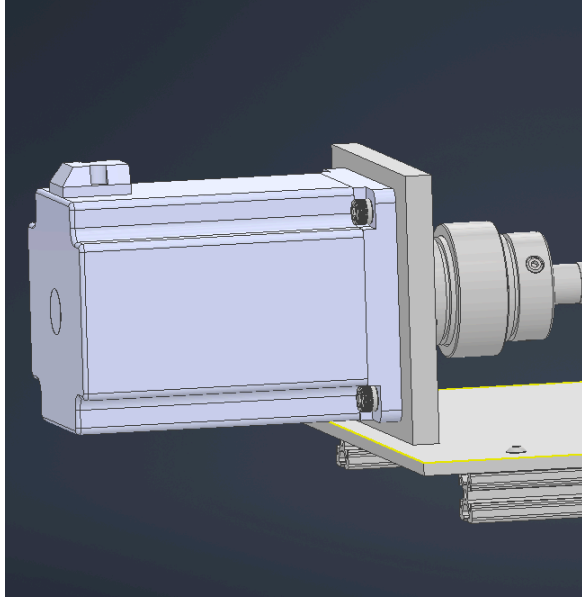


Figure 10. Motor Plate with motor attached

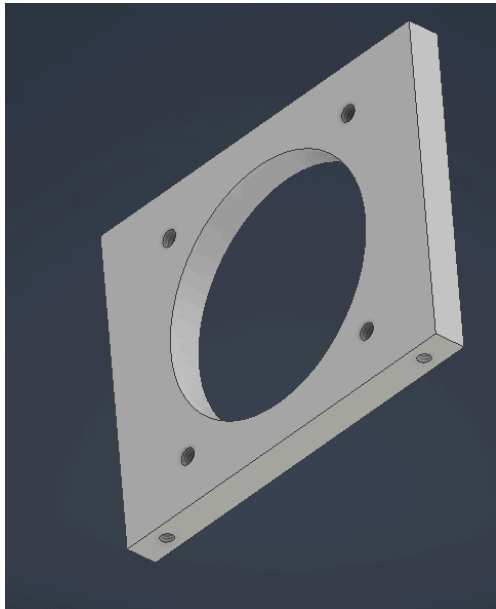


Figure 11. Motor Plate

Motor-Clutch Adapter

Since the motor shaft is 19mm in diameter and the clutch is designed for 20 mm shafts, a small adapter must be made to fill the gap. The adapter also has a slot for the key that will connect the motor to the armature of the clutch. **See Figure 12, 13.**

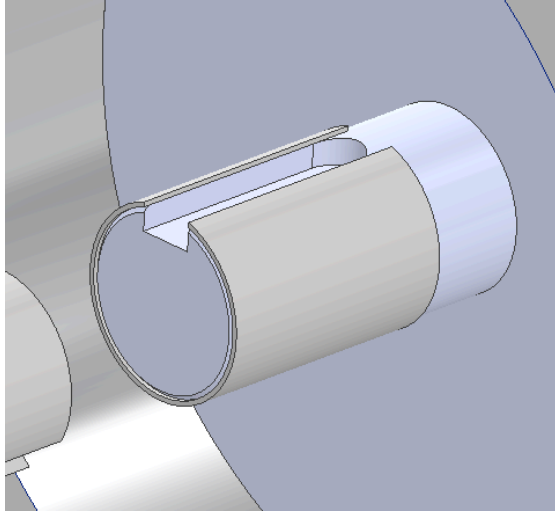


Figure 12. Motor-Clutch Adapter on motor shaft



Figure 13. Motor-Clutch Adapter

Driveshaft

The bulk part of the driveshaft is 20 mm in diameter to interface with the clutch properly. It has 2 6mm keyways, one on each side. The right side keyway is to the interface with the rotor of the clutch. The left side keyway interfaces with the shaft coupler. **See Figure 15.** The left end of the driveshaft is faced down to 19 mm so that a 19mm to 19mm shaft coupler can be used to connect the motor to the driveshaft. **See Figure 14.**



Figure 14. Shaft

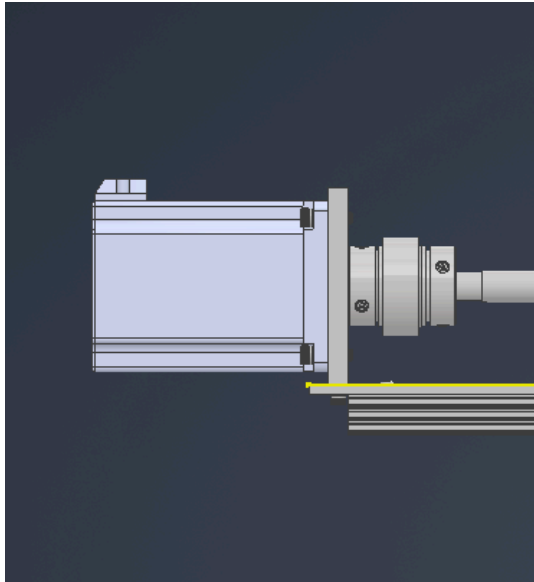


Figure 15. Shaft Coupler

Bearings and Bearing Risers

Since the driveshaft is rather long (~18 inches), the benchtop needs some other way to support the shaft along the length. 2 mounted bearings are used to ensure alignment and prevent deflection of the driveshaft. **See Figure 16.** To account for the added height of the motor, the mounted bearings are placed on top of 3D printed bearing risers. **See Figure 17.** The bearings and risers are bolted into the mounting plate via M8 bolts and nuts.

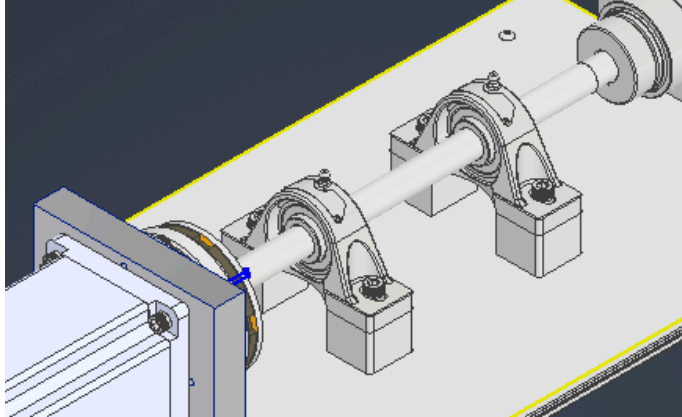


Figure 16. Mounted Bearings Isometric

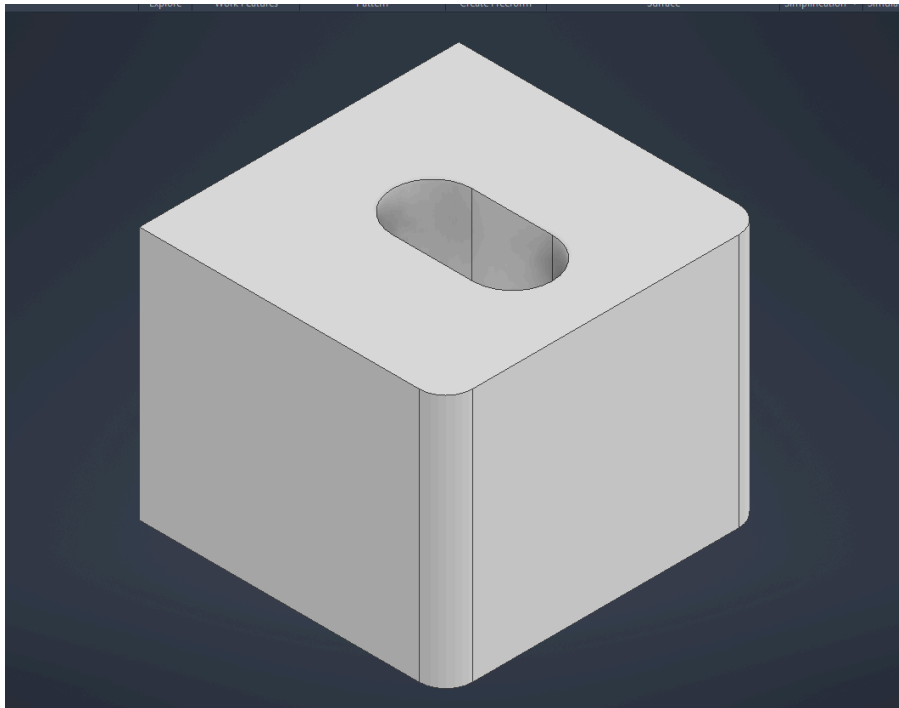


Figure 17. Bearing Riser

Main Structure

The main structure of the benchtop is composed of a $\frac{1}{4}$ " aluminum plate and 2 18" 80-20s. **See Figure 18, 19.** The mounting plate has holes for all major features, including the motor plate, clutch plate, and bearings. Additionally the plate has 4 mounting holes for M5 T-Slot hardware that are used to secure the 80-20s to the benchtop. **See Figure 20.**

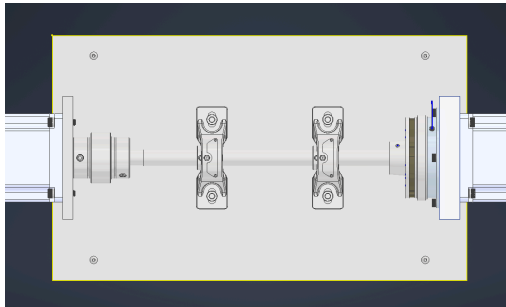


Figure 18. Benchtop Full Assembly Top Down View

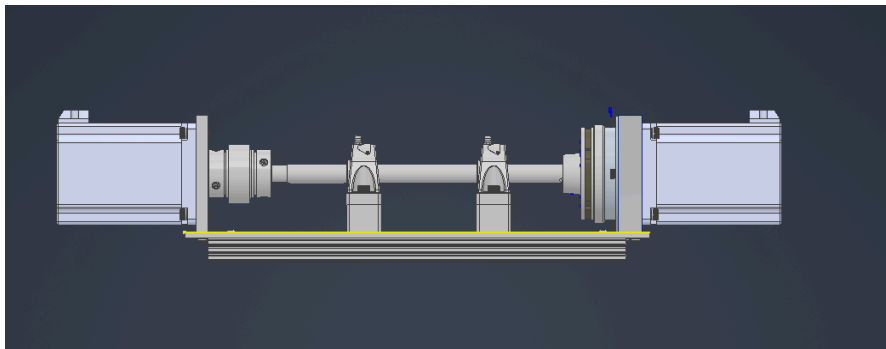


Figure 19. Mounted Bearings Side View

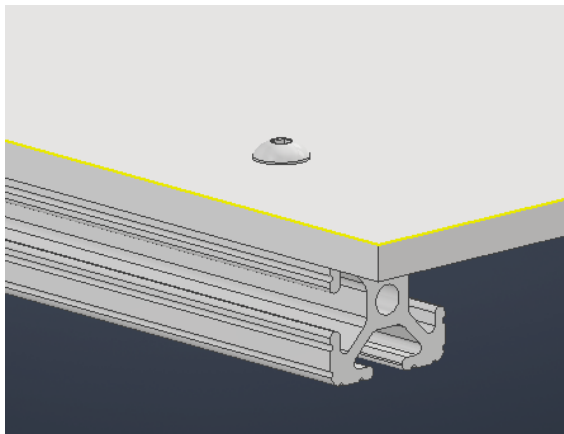


Figure 20. 80-20 Supporting Benchtop

Simulink

Aside from the clutch selection and the benchtop CAD, the largest portion of this project was writing the software to control the clutch. To do this, a Simulink block diagram was created. **See Figure 21.** The block diagram has 3 user inputs: an accelerator pedal, a forward/reverse switch, and a coast mode switch. With these 3 inputs, the user can select between forward coasting, forward locked, and reverse locked driving. Scopes are placed to measure the motor angular velocity, the load angular velocity, and the clutch torque. This block diagram can be divided into two parts: the dynamics and the controls.

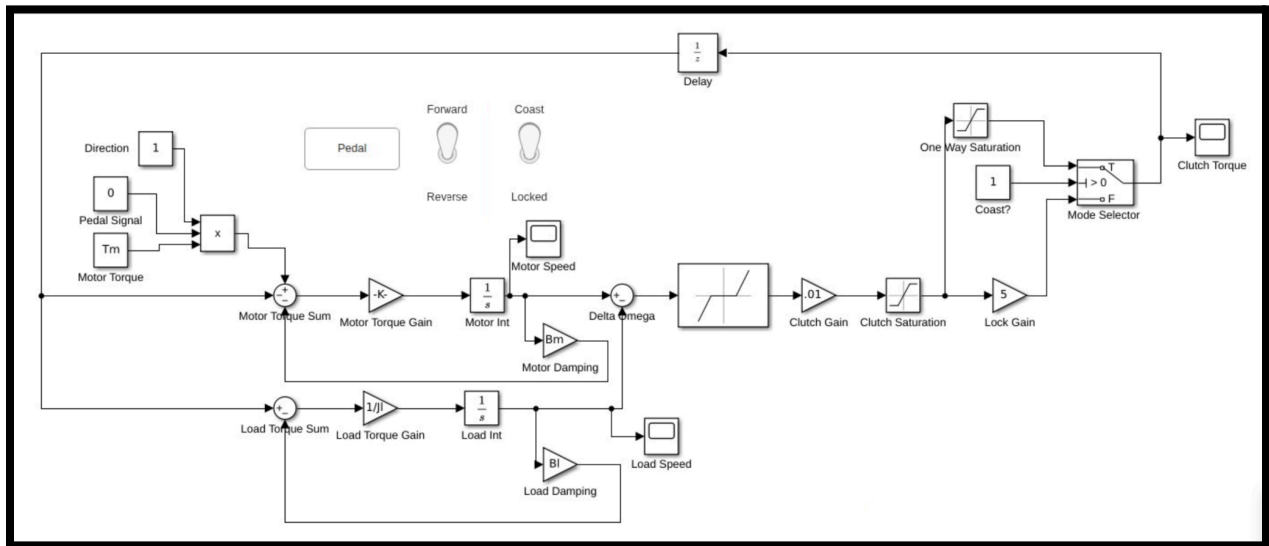


Figure 21.

Dynamics Model: Motor and Load

The physics of the motor and load are controlled by the following set of coupled differential equations.

Motor:

$$J_m \dot{\omega}_m = \tau_m - \tau_c - B_m \omega_m$$

Load:

$$J_l \dot{\omega}_l = \tau_c - B_l \omega_l$$

These are represented by the dotted green box, which turns sums of torques into angular velocities. **See Figure 22.** As shown in the equations and in the diagram, the motor angular velocity is affected by the torque from the motor, the torque from the clutch, and an additional damping term. The damping term accounts for frictional losses in the motor itself. Without this the motor would accelerate up to a constant speed and then rotate at that speed forever. The load angular velocity is affected by the torque from the clutch and its own independent damping term.

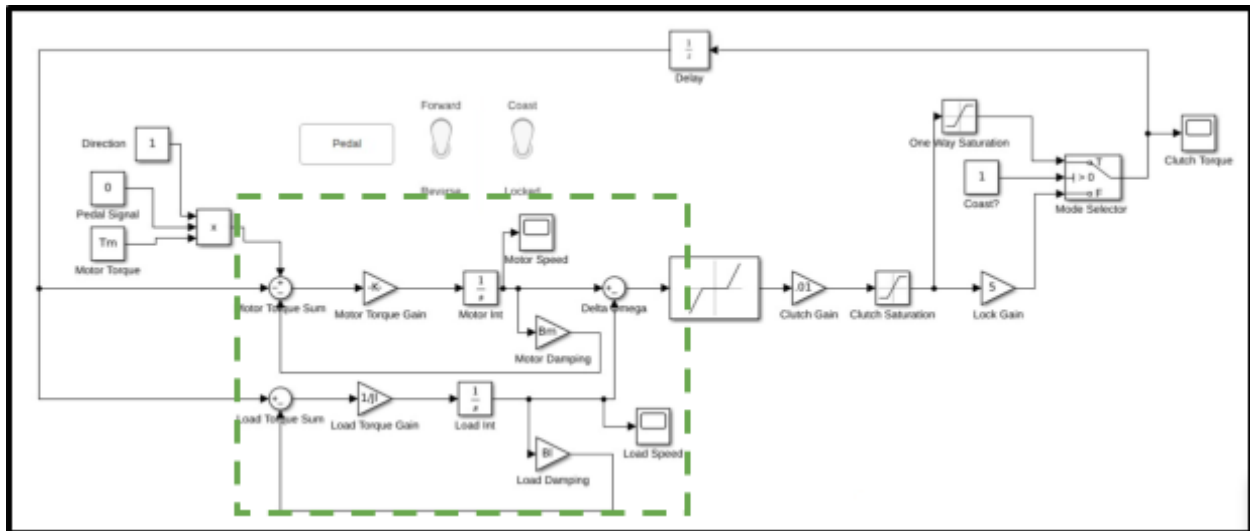


Figure 22. Block Diagram, Motor and Load Dynamics

Dynamics Model: Clutch

The block diagram starts off by calculating the raw torque. The $\Delta\omega$ signal goes through a deadzone block, which prevents chatter by only allowing clutch torque to be passed if a large enough $\Delta\omega$ is achieved. Then the new filtered $\Delta\omega$ is multiplied by K_c and passed through a saturation block to keep the clutch torque below its rated torque. At this point the signal splits to account for the two different operating modes. The top path is for coasting and the bottom is for a locked clutch. The top path passes the raw torque through another saturation block, which ensures that the torque is only positive. This is the result of the following two equations.

$$\tau_c = K_c \Delta\omega$$

$$\text{State} = \begin{cases} \text{Engaged,} & \omega_{\text{motor}} > \omega_{\text{wheel}} \\ \text{Disengaged,} & \omega_{\text{motor}} < \omega_{\text{wheel}} \end{cases}$$

Putting these two equations together, we see that with a positive gain K_c , the torque of the clutch can only ever be positive. A negative torque would have to come from $\Delta\omega$, but in this state, the clutch is disengaged, and the torque drops to zero. In the bottom path, the raw torque is multiplied by another gain, called the lock gain. The lock gain enforces the locking condition by artificially increasing the torque that the clutch transmits. This decreases the time it takes for the motor and load to synchronise. Lastly, a mode selector switch determines which torque is passed on to the rest of the loop. **See Figure 23.**

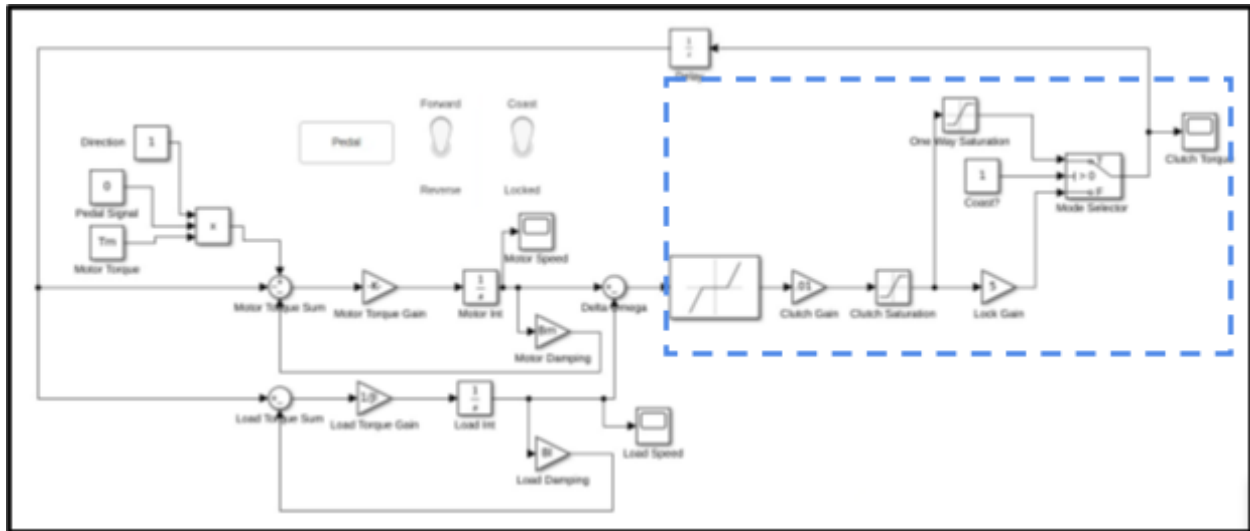


Figure 23. Block Diagram, Clutch Dynamics

Controls

Acceleration

The controls of this block diagram are integrated using dashboard buttons and toggle switches, which change the value of other variables elsewhere in the diagram. The pedal is a momentary switch that changes the pedal signal from 0 to 1. The direction switch is a bistable switch that changes the direction block from 1 to -1. Both of these values are fed into a product block, along with the motor's maximum torque, to create the motor torque. **See Figure 24.**

One Way Mode Selection

The coast/locked switch is also a bistable switch. It controls the “Coast?” block located on the right side of the block diagram. When “locked” is selected, the value of “Coast?” is switched to -1, which then causes the locked torque to be passed instead of the one-way saturated torque. **See Figure 24.**

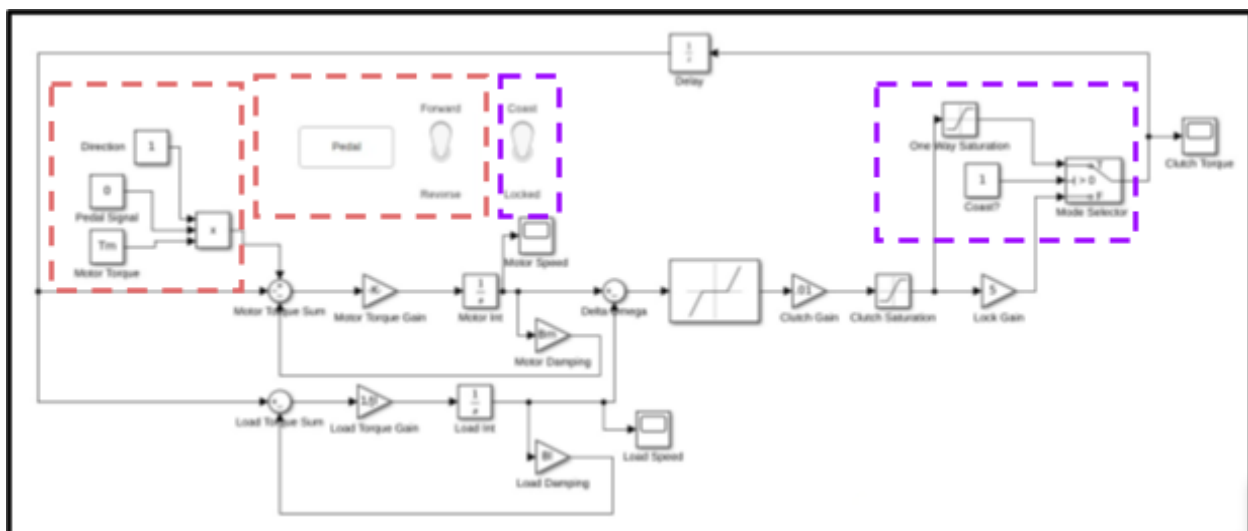


Figure 24. Block Diagram, Acceleration (red) and coast Mode Selection (purple)

Analysis

Simulation Verification

Plots of motor speed, load speed, and clutch torque were generated for each operating mode. During these simulations, the motor went through 2 throttle and release cycles. The forward coasting plots look just as expected; the motor speeds up the load, then when the pedal is released, the motor slows down while the load continues to coast. Additionally, the clutch torque is always positive, verifying that this clutch is functioning as a one way bearing. **See Figure 25.**

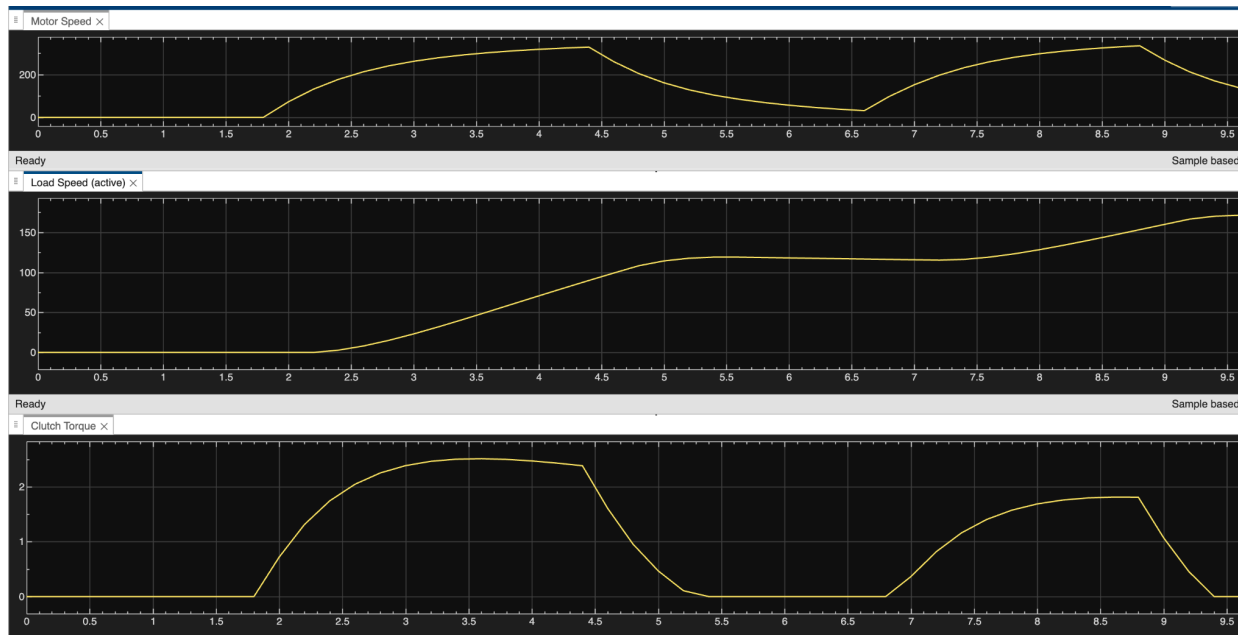


Figure 25. Forward Coasting Plots

In the forward locked plots, we see the opposite. The load speed increases and decreases with the motor speed. They are locked together. Additionally, the torque is positive or negative depending on the sign of $\Delta\omega$. **See Figure 26.** The results for the reverse locked plots look similar, but they are inverted. **See Figure 27.**

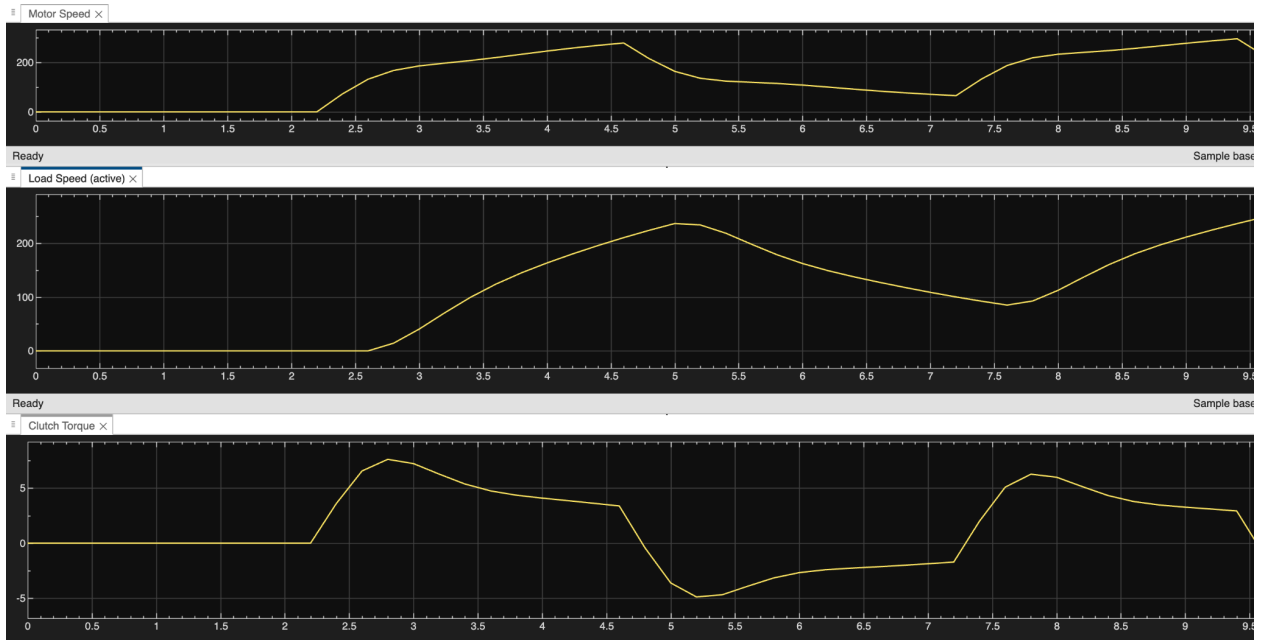


Figure 26. Forward Locked Plots

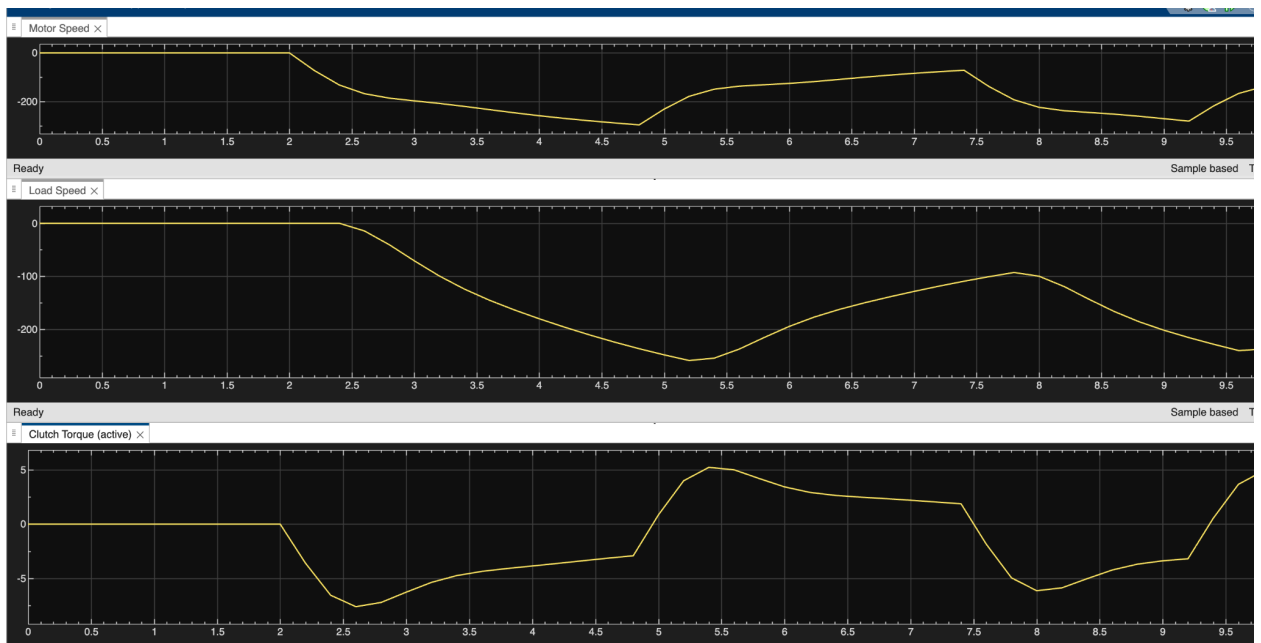


Figure 27. Reverse Locked Plots

Efficiency Analysis

For this design to be viable, the car's efficiency cannot decrease by more than 10% of its current value (37 mi/kWh). To calculate this drop in efficiency, there are two main factors: clutch power draw and added weight. Efficiency for electric vehicles is defined by the following formula.

$$\eta = \frac{\text{Total Distance Travelled}}{\text{Total Energy Consumed}}$$

In the Shell Eco-marathon, the distance travelled is fixed at 10 miles. Using this fact, we only need to know the total energy consumed to calculate the efficiency. The additional energy consumed by the clutch can be calculated using

$$E = P * t$$

where P is the rated power of the clutch and t is the time that the clutch is engaged throughout the whole race (time spent throttling). From historical data, the throttling time is typically only around 60% of the total race. Using a total race time of 35 minutes (the maximum allowable time) and a rated power of 20W, the total energy consumption of the clutch is calculated.

$$E = (20W)(.60)(35 \text{ min})(\frac{60 \text{ s}}{\text{min}}) = 25.2 \text{ kJ}$$

In order to compute the change in efficiency, the current energy consumption of the entire vehicle must be calculated. This can be done by converting the current vehicle's efficiency, 37 mi/kWh, into energy per distance, then multiplying by the 10 miles travelled in the race.

$$((\frac{37 \text{ mi}}{\text{kWh}}) * (\frac{\text{h}}{3600 \text{ s}}) * (\frac{\text{W*s}}{\text{J}}) * (\frac{1}{10 \text{ mi}}))^{-1} = 972.97 \text{ kJ}$$

With a fixed distance, calculating the percentage change in efficiency is as simple as taking a ratio of the total energy consumed in either case.

$$\frac{\eta_{\text{clutch}}}{\eta_0} = (\frac{972.97 \text{ kJ}}{972.97 \text{ kJ} + 25.2 \text{ kJ}}) = .974 \text{ or } 97.4\%$$

So the power draw of the clutch alone only decreases the total efficiency by 2.5%, or by around 1kWh. This is extremely small.

The other main method of efficiency loss is in the added mass of the clutch and the clutch mounting plate. To calculate the efficiency loss, the total energy consumed by the car over the entire race was calculated and compared to the total energy consumed by the car with the added mass. However, this calculation was done differently than in the previous section. In order to find the mass dependance of the efficiency, the total energy consumption was defined as the sum of energy lost in acceleration, rolling resistance, and aerodynamic drag. Furthermore, energy consumption is broken down into cycles of acceleration and coasting. The total energy is calculated by multiplying the energy per cycle by the number of cycles. The following equations were used to calculate the energy

$$E_{cycle} = E_{Acceleration} + E_{Rolling Resistance} + E_{Aero}$$

$$E_{cycle} = \frac{1}{2}m(v_{max} - v_{min})^2 + mg\mu D + \frac{1}{2}\rho C_d A v_{avg}^2 D$$

$$E_{total} = N * E_{cycle}$$

where m is the mass of the car with the driver, v_{max} is the maximum speed achieved during throttling, v_{min} is the minimum speed reached during coasting before accelerating again, g is the gravitational constant, μ is the coefficient of rolling resistance, D is the distance of one cycle, ρ is the density of air, C_d is the coefficient of drag, A is the frontal area of the car, v_{avg} is the average speed of the car, and N is the total number of cycles throughout the race.

The mass of the clutch is 1.66 kg and the mass of the clutch plate is 1.792 lbs (.812 kg), combining for a total mass of 3.452 kg. The mass of the clutch plate was calculated by applying material properties (Aluminum 6061) to the part in Autodesk Inventor. **See Figure 28.** The results for total energy consumed were calculated for the car with and without the clutch. The fraction of original efficiency is calculated by creating a ratio of these energies. **See Figure 29.** Due to the added mass, the efficiency only decreased by 1.5%, or by around .5 mi/kWh.

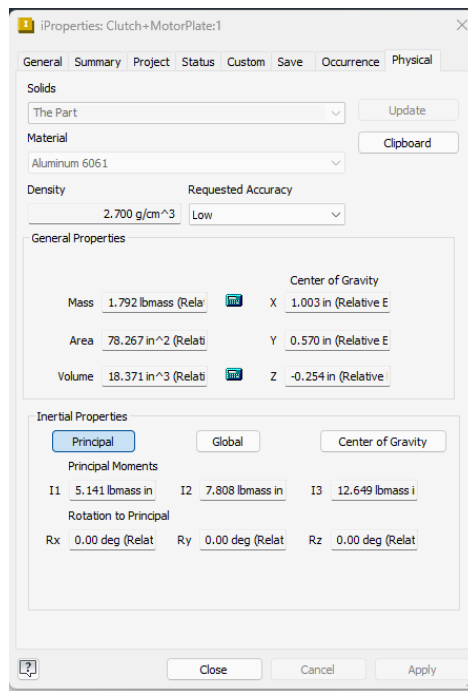


Figure 28. Mass of Clutch Plate

Number of Cycles	60 (15 per lap)			
Vmin	5.36448 m/s	Vavg	8.71729 m/s	
Vmax	12.0701 m/s			
Rolling Resistance	0.0236			
Distance of Cruising	268.1666667 m			
Mass	213.8688025 kg			
g	9.81 m/s ²			
Frontal Area	0.92525492 m ²			
Coefficient of Drag	0.153			
Density of Air	1.225 kg/m ³			
Per Cycle				
Accel	12501.66538 J/cycle			
Rolling Resistance	13278.01765 J/cycle			
Aero	1766.959961 J/cycle			
Total Energy per Cycle	27546.643 J/cycle			
Total Energy over Race	1652798.58 J			
D/E	0.0000060503440			
Number of Cycles	60 (15 per lap)			
Vmin	5.36448 m/s	Vavg	8.71729 m/s	
Vmax	12.0701 m/s			
Rolling Resistance	0.0236			
Distance of Cruising	268.1666667 m			
Mass	217.3208025 kg	Mass Increase	3.452 kg	
g	9.81 m/s ²			
Frontal Area	0.92525492 m ²			
Coefficient of Drag	0.153			
Density of Air	1.225 kg/m ³			
Per Cycle				
Accel	12703.45147 J/cycle			
Rolling Resistance	13492.33464 J/cycle			
Aero	1766.959961 J/cycle			
Total Energy per Cycle	27962.74607 J/cycle			
Total Energy over Race	1677764.764 J			
D/E	0.0000059603111:			
Percentage of Original Efficiency	0.9851193773			

Figure 28. Efficiency Loss Calculations for Added Mass

The total efficiency loss can be calculated by multiplying the individual efficiencies together. The result of this is an overall efficiency that is only 3.96% lower than previously, which is well within the 10% constraint set for this project. This corresponds to an efficiency of 35.5 mi/kWh, compared to the 37 mi/kWh from competition. This is a very small change.

Bill of Materials

BOM						
Part No.	Part Name/Desc	Vendor	Vendor Part No.	Unit Cost	Quantity	Part(s) Cost
1	Shaft Coupler	McMaster Carr	3565N1	\$67.03	2	\$134.06
2	Shaft Coupler Spider	McMaster Carr	3565N41	\$21.19	1	\$21.19
3	Shaft Coupler Spider Cover	McMaster Carr	3565N61	\$38.65	1	\$38.65
4	Bearings	McMaster Carr	7835T51	\$191.56	2	\$383.12
5	EM Clutch	McMaster Carr	101-10-11G-24V		1	0
6	Shaft	McMaster Carr	4634T42	\$31.67	1	\$31.67
7	Key Stock	McMaster Carr	92288A730	\$3.81	1	\$3.81
8	Motor	McMaster Carr	PEACO-D110BLD1000-30S	\$402.83	2	\$805.66
9	Shim Stock	McMaster Carr	95055K25	\$48.20	1	\$48.20
10	Motor Plate Stock	McMaster Carr	8975K217	\$24.79	1	\$24.79
11	Clutch Plate Stock	McMaster Carr	8975K352	\$40.16	1	\$40.16
12	Plate Stock	McMaster Carr	9246K425	\$119.04	1	\$119.04
13	M7 Bolts	McMaster Carr	91290A330	\$10.83	1	\$10.83
14	5/16" Bolts	McMaster Carr	91251A583	\$15.30	1	\$15.30
15	M8 Bolts	McMaster Carr	91290A458	\$11.07	1	\$11.07
16	M8 Nuts	McMaster Carr	90591A161	\$7.53	1	\$7.53
17	1/4-20 Bolts	McMaster Carr	91251A542	\$12.36	1	\$12.36
18	80-20s	McMaster Carr	47065T101	\$15.90	1	\$15.90
19	80-20 Hardware	McMaster Carr	3136N409	\$5.46	1	\$5.46
					Total Cost=	\$1,728.80

Note: The price of the clutch is not included in the BOM since the price was not listed. Additionally, the cost of the motors is included for completeness, but the team already owns these motors.

Manufacturing Plan

Manufacturing Plan		
Part	Method	Material
Shaft	1) Cut to size with drop saw 2) Decrease from 20mm to 19 mm with lathe 3) Make keyways with mill and a collet block	Aluminum 6061
Motor-Clutch Adapter	Bend shim stock into cylinder (potentially hydraulic press with mold)	Aluminum 6061
Motor Plate	CNC, then tap holes manually	Aluminum 6061
Clutch Plate	CNC, then tap holes manually	Aluminum 6061
Mounting Plate	Mark holes, then use drill press	Aluminum 6061
Bearing Riser	3D print	PLA

Project Reflection and Future Work

<If you were to start this project over, what would you change about the design and your process?

If I were to start this project again, I would try to make a physical assembly as soon as possible. With this physical benchtop, it would be much easier to write the controls, since you do not need to model all of the physics as well. The physical benchtop would also be more accurate than the physical model built in Simulink. Although my simulation does behave as expected, that does not guarantee that this control logic will work in real life. That is why testing is so important. Testing often reveals problems that you didn't even know existed.

Design wise, I think there is a lot of potential for improvement in terms of sensors and data collection. Right now the only available information is the rpm of the drive motor and the load motor, but a joulemeter could be useful to measure the energy consumption of the motor and the clutch. An odometer could also be integrated in software so that we can measure the distance traveled. These two data points can be used in tandem to calculate a more realistic efficiency of the motor-clutch combination. It accounts for mechanical losses of slip that my calculations do not account for, while adding information about how efficiently the motor turns electrical energy into mechanical energy. Additionally, with the proper analysis, the clutch plate mass can be cut down quite a bit. Lastly, more consideration should be given to how the motor and clutch will be mounted in the real car.

In the future, this drivetrain configuration has the potential to be used for regenerative braking. This is perhaps the most exciting aspect of this project. Because the motor is not directly connected to the wheels, the opportunity to coast still exists. In this architecture, the car could accelerate and coast most of the race, which has been deemed the most efficient method. Regenerative braking could be used to slow the car down for a mandatory stop. That way, the energy lost in braking can be minimized.

Appendix

A.1. Alternative Design Decision Matrix

Category Weights	0.2	0.1	0.1	0.05	0.4	0.15	1
Designs	Weight (lowest)	Mechanical Complexity (lowest)	Power Draw (lowest)	Software Complexity (lowest)	Time to switch between modes (lowest)	Space (lowest)	Total
Software Enhanced Mechanical Clutch	3	3	3	2	5	3	3.75
Software Enhanced EM Clutch	4	4	4	2	5	5	4.45
Modular Wheels	5	5	5	5	1	5	3.4
Double Clutch Transmission	1	1	5	5	5	1	3.2

A.2 Clutch Testing Benchtop CAD Assembly

File location in CEV Vault: Designs/!Sandbox/JVogel/Senior Design/ClutchDynoAss.iam

A.3 Efficiency Calculations Spreadsheet

Clutch Power	20 W	Number of Cycles	60 (15 per lap)		
Percent of Race Throttling	60.00%	Vmin	5.36448 m/s	Vavg	8.71729 m/s
Length of Race	10 miles	Vmax	12.0701 m/s		
Time of Race	35 min	Rolling Resistance	0.0236		
Time of Race using Clutch	21 min	Distance of Cruising	268.166667 m		
Clutch Energy Consumption (Clutch)	25200 J	Mass	213.8688025 kg		
		g	9.81 m/s^2		
		Frontal Area	0.92525492 m^2		
		Coefficient of Drag	0.153		
Efficiency Before	37 mi/kWh	Density of Air	1.225 kg/m^3		
Total Energy Consumption	972972.973 J				
Percent	1.0259	Per Cycle			
Efficiency Relative to Before	97.48%	Accel	12501.66538 J/cycle		
Percent change	2.52%	Rolling Resistance	13278.01765 J/cycle		
Actual Number	36.06589336	Aero	1766.959961 J/cycle		
		Total Energy per Cycle	27546.643 J/cycle		
		Total Energy over Race	1652798.58 J		
		D/E	0.0000060503440		
		Number of Cycles	60 (15 per lap)		
		Vmin	5.36448 m/s	Vavg	8.71729 m/s
		Vmax	12.0701 m/s		
		Rolling Resistance	0.0236		
		Distance of Cruising	268.166667 m		
		Mass	217.3208025 kg	Mass Increase	3.452 kg
		g	9.81 m/s^2		
		Frontal Area	0.92525492 m^2		
		Coefficient of Drag	0.153		
		Density of Air	1.225 kg/m^3		
		Per Cycle			
		Accel	12703.45147 J/cycle		
		Rolling Resistance	13492.33464 J/cycle		
		Aero	1766.959961 J/cycle		
		Total Energy per Cycle	27962.74607 J/cycle		
		Total Energy over Race	1677764.764 J		
		D/E	0.0000059603111		
		Percentage of Original Efficiency	0.9851193773		
Fraction of Original Efficiency	0.9602489301				
Percentage of Efficiency Lost	0.03975106994				
Corresponds to a loss of	1.470789588 mi/kWh				
or a new total of	35.52921041 mi/kWh				

A.4 Bill of Materials

BOM						
Part No.	Part Name/Desc	Vendor	Vendor Part No.	Unit Cost	Quantity	Part(s) Cost
1	Shaft Coupler	McMaster Carr	3565N1	\$67.03	2	\$134.06
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